Introduction to Numerical Analysis

Lecture 1

Motivation

- Can you solve analytically?

$$e^x = x^2$$

Motivation

- Can you solve analytically?

$$e^x = x^2$$

  - Many real-world problems have no closed-form solution
  - Computers solve them numerically

Motivation

Solution of Nonlinear Equations

- Some simple equations can be solved analytically:

$$x^2 + 4x + 3 = 0$$

$$\text{Analytic solution } roots = \frac{-4 \pm \sqrt{4^2 - 4(1)(3)}}{2(1)}$$

$$x = -1 \text{ and } x = -3$$

- Many other equations have no analytical solution:

$$\left.\begin{array}{c} x^9 - 2x^2 + 5 = 0 \\ x = e^{-x} \end{array}\right\} \text{No analytic solution}$$

Motivation

Solution of Systems of Linear Equations

$$x_1 + x_2 = 3$$

$$x_1 + 2x_2 = 5$$

We can solve it as :

$$x_1 = 3 - x_2, \qquad 3 - x_2 + 2x_2 = 5$$

$$\Rightarrow x_2 = 2,\ x_1 = 3 - 2 = 1$$

What to do if we have

1000 equations in 1000 unknowns.

Motivation

- How does AI train models?
- How does a control system compute responses?
- How do we simulate circuits?
  - All rely on **numerical computation**.

What is Numerical Analysis (NA)?

- **Numerical Analysis** is a branch of mathematics and computer science that focuses on:
  - Designing **algorithms**
  - Developing **approximation methods**
  - Analyzing **numerical errors**
  - Implementing solutions using **computers**

  to solve **mathematical problems that cannot be solved exactly or efficiently by analytical methods**.

What is Numerical Analysis?

- Study of algorithms for numerical approximation
  - Focuses on **accuracy**
  - Efficiency
  - Stability
  - Error behavior

Numerical Analysis = Mathematics + Algorithms + Computing

Why Do We Need Numerical Analysis?

- Many real-world problems:
  - Do not have exact (closed-form) solutions
  - Are too complicated to solve symbolically
  - Involve very large systems
  - Require fast computation

Why Do We Need Numerical Analysis?

- Examples:
  - Solving nonlinear equations like $e^x = x^2$
  - Solving large systems of linear equations (thousands or millions of variables)
  - Computing integrals that have no exact formula
  - Simulating physical systems (heat transfer, control systems, AI models)

Applications of Numerical Analysis



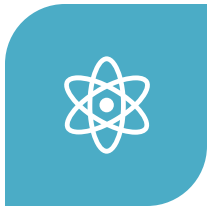
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CONTROL ENGINEERING



FINANCE



SCIENTIFIC COMPUTING



ELECTRICAL ENGINEERING



MECHANICAL ENGINEERING



CLIMATE MODELING



COMPUTER GRAPHICS

Topics in This Course

- Taylor Theorem
- Solution of nonlinear Equations
- Solution of linear Equations
- Least Squares curve fitting
- Interpolation
- Numerical Differentiation
- Numerical Integration
- Solution of ordinary differential equations
- Solution of Partial differential equations

Reference book



Numerical Methods for Engineers, 6th edition by Steven C. Chapra and Raymond P. Canale

Methods for Solving Nonlinear Equations

- **Bisection Method**
- **Method of False Position**
- **Fixed-Point Iteration**
- **Newton-Raphson Method**
- **Secant Method**

Methods for Solving linear Equations

- Naive Gaussian Elimination
- Gaussian Elimination with ScaledPartial Pivoting
- LU-Decomposition
- Inverse of a Matrix
- Gauss-Seidel Methods

Methods for Curve Fitting

Least Squares

1. Linear Regression
2. Nonlinear Least Squares Problems



Interpolation

- Newton Polynomial Interpolation
- Lagrange Interpolation
- Quadratic Splines

Methods for Numerical Integration

- Trapezoid Method
- Simpson's Rule
- Gauss Quadrature

- Some algorithms are:
  - Accurate but slow
  - Fast but unstable
  - Stable but memory expensive
- We must choose the best trade-off.
- Numerical Analysis connects:
  - Mathematics
  - Algorithms
  - Hardware limitations
  - Computational complexity

Course Objective

- After this course, students should be able to:
  - Analyze numerical errors
  - Design numerical algorithms
  - Compare different numerical methods
  - Implement algorithms using Python or MATLAB
  - Evaluate stability and convergence
  - Apply NA in engineering and AI problems

MATLAB or Python

- MATLAB is ideal if:
- Highly optimized for matrix operations
  - Focus is purely on numerical algorithms
  - Students are not strong programmers
  - Course is math-oriented
- Python is ideal if:
- Comparable performance using NumPy
  - You want integration with:
    - Machine Learning
    - Data science
    - Scientific computing
  - You want students to build real-world skills
  - You want a free solution

Exact vs Numerical Solutions

- Exact → symbolic formula
- Numerical → approximate value

  – In real engineering problems, we usually accept highly accurate approximations instead of exact (true) solutions.

Error Analysis

- Absolute error
- Relative error
- Round-off error
- Truncation error

Error Analysis

“Is error always bad?”

Error Analysis

- “Is error always bad?”
  - No.
- Error is acceptable if:
  - Controlled
  - Bounded
  - Does not grow

Absolute Error

- The relationship between the exact, or true, result and the approximation can be formulated as
  - true value = approximation + error
- the numerical absolute true error is equal to
  - $E_t = |\text{true value} - \text{approximation}|$

Relative Error

- Can be computed if the true value is known:

Absolute True Error

$$E_t = \left| \text{true value} - \text{approximation} \right|$$

Absolute Percent Relative Error

$$\varepsilon_t = \left| \frac{\text{true value} - \text{approximation}}{\text{true value}} \right| * 100$$

Estimated Error

- When the true value is not known:

Estimated Absolute Error

$$E_a = \left| \text{current estimate} - \text{previous estimate} \right|$$

Estimated Absolute Percent Relative Error

$$\varepsilon_a = \left| \frac{\text{current estimate} - \text{previous estimate}}{\text{current estimate}} \right| * 100$$

We say that the estimate is correct to *n* decimal digits if:

$$\left|\text{Error}\right| \leq 10^{-n}$$

We say that the estimate is correct to *n* decimal digits **rounded** if:

$$\left|\text{Error}\right| \leq \frac{1}{2} \times 10^{-n}$$

Example

- Suppose that you have the task of measuring the lengths of a bridge and a rivet and come up with 9999 and 9 cm, respectively.
- If the true values are 10,000 and 10 cm, respectively, compute the true error the true percent relative error for each case.

Example

- Suppose that you have the task of measuring the lengths of a bridge and a rivet and come up with 9999 and 9 cm, respectively.
- If the true values are 10,000 and 10 cm, respectively, compute the true error the true percent relative error for each case.

- The true error for measuring the bridge is

$$E_t = 10{,}000 - 9999 = 1 \text{ cm}$$

and for the rivet it is

$$E_t = 10 - 9 = 1 \text{ cm}$$

- The percent relative error for the bridge is

$$\text{perecent } relative\ error\ (\varepsilon_t) = \left|\frac{10{,}000 - 9999}{10{,}000}\right| \times\ 100\% = 0.01\%$$

- The percent relative error for the rivet is

$$\text{perecent } relative\ error\ (\varepsilon_t) = \left|\frac{10 - 9}{10}\right| \times\ 100\% = 10\%$$

Sources of Error

Round of Error

Truncation Error

“Does the computer store real numbers exactly?”

Round-off errors

- Floating Point Representation
- x = ± m × 2^e
  - Finite precision storage
- Computers do not store real numbers exactly.
  Example: 1/3=0.333333……
  - The computer stores only a limited number of digits.
  - This leads to:
    - Round-off errors
    - Accumulated numerical errors

Round-off errors

- Numbers that have a finite expansion in one numbering system may have an infinite expansion in another numbering system:

$$(1.1)_{10} = (1.0001100110011001100...)_2$$

- You can never represent 1.1 exactly in binary system.

Truncation Error

- Occurs when approximating infinite processes
  - Example: Taylor series approximation

Loss of Significance

- Occurs when subtracting nearly equal numbers
  - Can cause large relative error

Convergence

- A method converges if approximations approach true value

$$\lim_{n \to \infty} x_n = x_{true}$$

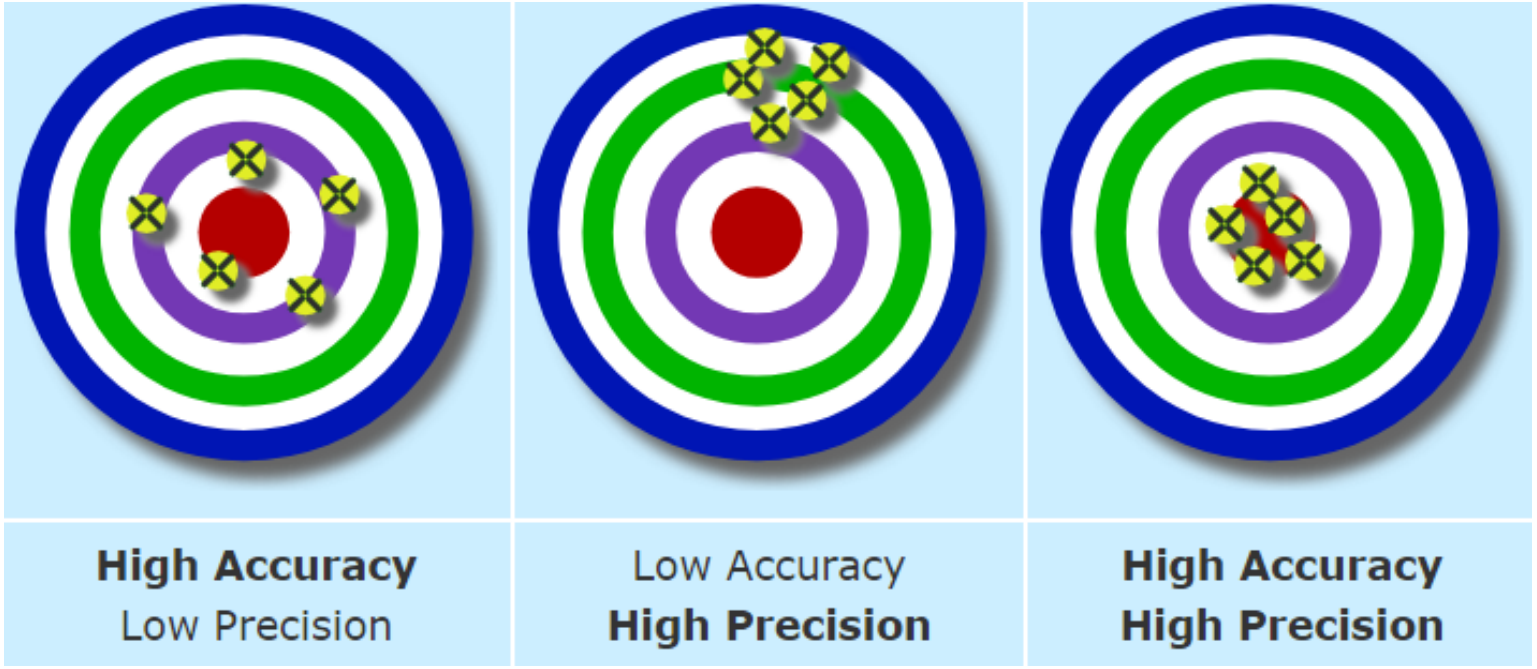
Stability

- Stable: small input error → small output error
- Unstable: error grows during computation
- A method can be:
  - Convergent but unstable
  - Stable but slow
  - Stable and fast (ideal)

Accuracy and precision

- The errors associated with both calculations and measurements can be characterized with regard to their accuracy and precision.
- ***Accuracy:*** refers to **how closely** a computed or measured value agrees with the true value.
- ***Precision:*** refers to how closely individual computed or measured values agree with each other.

Examples of Accuracy and Precision:



High Accuracy Low Precision	Low Accuracy **High Precision**	**High Accuracy** **High Precision**

Significant Figures

- Number of significant figures indicates precision.
- Significant digits of a number are those that can be used with confidence, e.g., the number of certain digits plus one estimated digit.
- Zeros are sometimes used to locate the decimal point not significant figures.

0.00001753	4
0.001753	4

Identifying Significant Digits

- All non-zero digits are considered significant:
  - For example, 91 has two significant figures, while 123.45 has five significant figures.

- Zeros appearing anywhere between two non-zero digits are significant:
  - For example: 101.1002 has seven significant figures.

- Leading zeros are not significant:
- For example: 0.00052 has two significant figures.

Identifying Significant Digits

- Trailing zeros in a number containing a decimal point are significant.
- For example:
  - 12.2300 has six significant figures: 1, 2, 2, 3, 0 and 0.
  - The number 0.000122300 still has only six significant figures.
  - In addition, 120.00 has five significant figures.

- The significance of trailing zeros in a number not containing a decimal point can be ambiguous.
- For example,
  - it may not always be clear if a number like 1300 is accurate to the nearest unit. Various conventions exist to address this issue.

Identifying Significant Digits

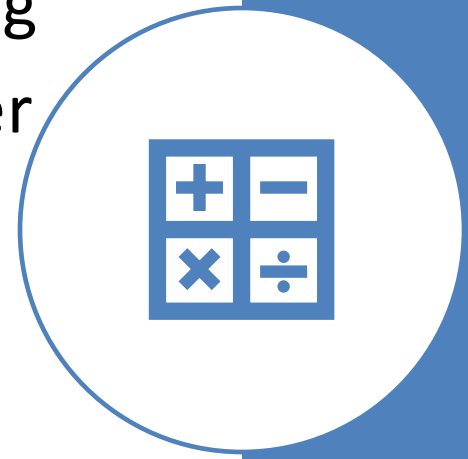
- How many significant digits of these numbers:

  3.14159
  0.00082
  0.000820
  3009
  250
  250.0
  and 1000?

Number	Number of significant digits	
3.14159	6	All digits are interesting
0.00082	2	only the 8 and 2 is interesting; the zeroes are placeholders; they may have rounded something off to get this value.
0.000820	3	only the 8, 2 and 0 tell us something; the other zeroes are placeholders
3009	4	the 3 and 9 are interesting, and we have to count the zeroes, because they're between the two interesting numbers.
250	2	the last zero is just a placeholder.
250.0	4	the zero in the tenths place means that the measurement was made accurate to the tenths place, and that there just happen to be zero tenths; the 2 and 5 give useful information, and the other zero is between significant digits, and must therefore also be counted
1000	1	only the 1 is interesting (only it tells us anything specific); we don't know anything for sure about the hundreds, tens, or units places; the zeroes may just be placeholders

Summary

- Numerical Analysis is:
  - The backbone of scientific computing
  - Essential for engineers and computer scientists
  - The foundation of AI, simulation, optimization, and modeling
  - The science of turning mathematics into computable algorithms



Key Message

- Numerical Analysis teaches you
  - **When to trust a computed result**